

Vidya360

AI-Powered Fee Intelligence, Academic Insights & Credit Layer — A 360° View of Every Student

Concept Document — Smart School FinTech Innovation Challenge 2026

1. Executive Summary

PaperBuddy and most school-ERP tools answer a narrow question: how much fee is due, and who hasn't paid. None of them answer the harder fintech questions that actually change outcomes for families and institutions: who is likely to default before it happens, how can payment be made manageable instead of punitive, what happens to a student's payment history when they change schools, and what does the institution's cash position look like next month.

Vidya360 is positioned as a fintech intelligence layer that plugs on top of any existing school ERP (including PaperBuddy) rather than replacing it. It is built around seven pillars. Three of them are the focus of this document's latest updates: a portable student record that follows a student across schools, a fee-payment 'credit score' (EduScore) that lets a new school assess a transferring student without demanding a large upfront payment, and a unified academic performance & achievements view that sits alongside the fee dashboard so the platform captures a student's whole journey, not just their payments.

2. The Market Gap

What PaperBuddy and peers do well: fee collection, reminders, receipts, basic reporting, parent-facing payment portals.

What none of them do: predict risk before it happens, offer flexible repayment tied to that risk, treat a student's payment history as a portable financial asset, or forecast institutional cash flow from fee data.

That gap is the fintech opportunity. Building "PaperBuddy with more features" competes on the sponsor's own turf and loses. Building the fintech layer they lack wins on differentiation and matches the hackathon's explicit "FinTech Innovation" framing.

3. Product Pillars — Overview

Vidya360 is built around seven pillars. The first two were part of the original concept; the following three are new additions covering school-transfer continuity, a portable credit score, and a unified academic view.

- **Defaulter Risk Scoring** — predicts which families are likely to miss a payment, before it happens.
- **Flexible Payment Plans (EMI / split-pay)** — auto-generated based on risk score, so lump-sum pressure is replaced with manageable installments.
- **FeePassport (NEW)** — a portable, verifiable financial + attendance-linked record that transfers with a student when they change schools.
- **Academic Performance & Achievements Dashboard (NEW)** — results, attendance trends, and co-curricular achievements shown alongside the fee view, and carried along with FeePassport on transfer.
- **EduScore (NEW)** — a CIBIL-like credit score for a student's family, built from fee-payment timeliness, that a new school can use to waive or reduce upfront payment at admission.
- **Cash-Flow & Runway Dashboard** — forecasts an institution's collectible revenue over the coming terms.
- **Unified Parent/Admin Dashboard** — single pane of glass for fees, results, achievements, and admin operations.

4. Defaulter Risk Scoring

A simple, explainable model — logistic regression or gradient boosting, not deep learning — scores each family's likelihood of a late or missed payment. Explainability matters more than raw accuracy in a demo and in front of school administrators who need to justify decisions to parents.

Signals used

- Historical on-time vs late payment ratio, per term
- Trend over time (improving, stable, deteriorating)
- Number of siblings and combined fee load per family
- Days-past-due distribution on past invoices
- Communication responsiveness (did the family respond to reminders)

Output

A 0–100 risk score per family per term, refreshed after every payment cycle, with a plain-language explanation generated alongside it (e.g. “Risk elevated due to two consecutive late payments and a new sibling enrollment”).

5. Flexible Payment Plans (EMI / Split-Pay)

Instead of forcing a termly lump sum, the system auto-generates a split-payment or EMI-style plan sized to the family's risk profile and payment history. This is the single largest unmet need in Indian schools: most families don't default out of unwillingness, they default because a lump sum is operationally hard to manage against irregular income.

- Low risk → standard due date, optional early-bird discount
- Medium risk → 2–3 installment split, auto-scheduled reminders
- High risk → monthly micro-installments, plus an optional counseling flag for the school admin (not a punitive flag — a support flag)

6. FeePassport — Portable Student Record (NEW)

Today, when a student changes schools, their fee-payment history, discount history, and financial standing don't travel with them. The old school has no reason to share it, the new school has no way to verify it, and the parent starts from zero — often facing a full upfront deposit purely because the new school has no data to trust them with anything less.

What FeePassport transfers

- Fee-payment timeline: dues raised, dues paid, dates, and any installment plans honored or missed
- Discount / scholarship history (with reason codes, not personal financial detail)
- Attendance-linked enrollment record (start date, end date, class/grade, reason for leaving if disclosed)
- The computed EduScore (see Section 8) at time of transfer, with its trend history
- Academic performance and achievements record (see Section 7), so the new school receives results and co-curricular history alongside financial history

How it works

- Parent-initiated, consent-based transfer: the parent (not the old school unilaterally) authorizes release of the record to the new school — this keeps the feature GDPR/DPDP-compliant and avoids it becoming a blacklist mechanism.
- Old school issues a cryptographically signed record (a JSON credential, verifiable without calling the old school directly) once the parent requests transfer.
- New school's Vidya360 instance verifies the signature and imports the record directly into its own database — no manual data entry, no phone calls between admin offices.
- If the old school isn't on Vidya360, a manual verification workflow (upload of last fee receipt + admin attestation) provides a fallback so the feature still works outside a closed network.

Why this matters for judging

Data portability is a real fintech pattern (think credit bureau data following a borrower between banks). Applying it to school transfers is a genuinely novel idea in this space and solves a real, common pain point for transferring families — particularly relevant in India given high inter-city relocation for work.

7. Academic Performance & Achievements Dashboard (NEW)

Correlates with: FeePassport (portability) + Unified Dashboard (surface). Today's parent dashboard shows only money — dues, EMIs, receipts. Parents and admissions teams also want the other half of the picture: how the student is actually doing academically and beyond the classroom. Folding that into the same dashboard — and the same portable record — turns Vidya360 from a fee tool into a genuine single source of truth for a student's school journey.

What the dashboard shows

- Term-wise results: subject-wise marks/grades, overall percentage or GPA, and a trend line across terms so improvement or decline is visible at a glance.
- Attendance percentage, shown alongside results since the two are naturally read together by parents and admins.
- Achievements & co-curriculars: sports, olympiads, competitions, arts, NCC/NSS, certifications — each with title, category, level (school / district / state / national), date, and an optional certificate/photo upload.
- Teacher remarks or badges per term, giving qualitative context next to the quantitative result.
- A single “student snapshot” card usable by admissions staff at a new school, combining this section with the EduScore/FeePassport summary — one screen instead of three separate conversations with the old school.

How it's built

Layer	What it needs	Reuse
Data entry	Result upload (CSV/manual) per term; achievement entry form with file upload	Reuses existing student/term structure from Section 11
Storage	New AcademicRecord and Achievement entities (see updated data model)	New tables, same database and student key
Surface	New tab/section on the existing parent and admin dashboards	Reuses dashboard shell and components built for fee views
Portability	Bundled into the FeePassport payload on transfer	Full reuse of Section 6's signing/verification flow

Important boundary

Academic performance is deliberately kept separate from EduScore and the defaulter risk score. EduScore stays a narrow, fee-payment-reliability score; academic results are shown alongside it for context, never blended into it. Mixing the two would let a family's financial score be affected by a child's grades (or vice versa), which is neither fair nor something a school should be able to justify to a parent — keeping them visually adjacent but functionally separate protects the credibility of both.

8. EduScore — Student Fee Credit Score (NEW)

EduScore is a CIBIL-style score, but scoped narrowly to fee-payment reliability rather than general creditworthiness. A new school can use it at admission time to decide whether to waive or reduce the upfront deposit that would otherwise be demanded from an unknown family.

Score composition (illustrative weighting)

Factor	Weight	What it captures
On-time payment ratio (last 3 years)	40%	Proportion of dues paid by original due date
Recency of any missed payment	20%	Recent lapses weigh more than old ones
Consistency across siblings/family	15%	Whether the pattern holds across all children in the family
Response to reminders/plans	15%	Whether the family engaged and resolved dues, even if late
Tenure / data depth	10%	More terms of history → more confidence in the score

Score bands and admission impact

Band	Score range	Suggested new-school action
Excellent	750–900	No upfront deposit required; standard term billing from day one
Good	650–749	Reduced deposit (e.g. one month instead of full term)
Fair	500–649	Standard deposit, but EMI plan offered automatically
Building history	No score / new family	Standard deposit; score begins accruing from term one

Important safeguards

- This is an enabling score, not a punitive one: its only sanctioned use is to reduce barriers (lower/waived deposits) for reliable families. It must never be used to deny admission outright — that reframes a fintech convenience feature into a discriminatory gatekeeping tool, which is both an ethical and a legal risk.
- A family with no score (new to the system) is never treated as “high risk” by default — absence of data is neutral, not negative.
- Score and underlying payment history are visible to the family themselves at any time, mirroring RBI/credit-bureau transparency norms for consumer credit scores.
- Disputes: a simple flow lets a parent flag an incorrect due/paid record for correction before it affects the score.

9. Cash-Flow & Runway Dashboard

Aggregates individual risk scores and payment-plan schedules into an institution-level forecast: expected collections for the next 1–3 terms, expected shortfall from at-risk families, and a simple “runway” view for school finance staff. This turns a pile of individual predictions into a single number a principal or trustee actually cares about.

10. Unified Parent / Admin Dashboard

- Parent view: dues, EMI schedule, EduScore with plain-language explanation, term-wise results and attendance trend, achievements timeline, and a one-tap request to transfer FeePassport (fee + academic record) to a new school.
- Admin view: risk-scored fee pipeline (Kanban-style), cash-flow forecast, class-wise academic performance summary, and an inbound-transfer inbox for verifying incoming FeePassport records from other schools.

11. Updated Data Model (Prisma-style, high level)

Entity	Key fields	Notes
Student	id, name, guardianId, currentSchoolId, enrollmentHistory[]	enrollmentHistory supports multi-school tenure
Guardian	id, name, contact, familyId	familyId groups siblings for combined risk signal
FeeInvoice	id, studentId, termId, amountDue, dueDate, status	status: paid / partial / overdue / plan-active
Payment	id, invoiceId, amount, paidDate, method	feeds both risk score and EduScore
RiskScore	id, familyId, termId, score, explanation, computedAt	recomputed per cycle
EduScore	id, familyId, score, band, history[], lastUpdated	carried inside FeePassport on transfer
FeePassportRecord	id, studentId, issuingSchoolId, payload (signed), issuedAt	the exportable/importable credential
PaymentPlan	id, invoiceId, installments[], riskBand	auto-generated EMI schedule
AcademicRecord	id, studentId, termId, subject, marks/grade, attendancePercent	feeds results view; bundled into FeePassport
Achievement	id, studentId, title, category, level, date, certificateUrl	co-curricular/achievement entries, shown on dashboard

12. Suggested Hackathon Build Order (48–72h format)

Hours	What to build	Notes
0–6	Core schema: students, guardians, fee heads, payments, risk & EduScore tables	Foundation for everything else
6–18	Payment gateway test-mode flow + webhook reconciliation	Standard, keep it simple and reliable
18–28	Risk scoring service (heuristic or lightweight model) + plain-language explanation	Explainability over model complexity
28–38	Auto EMI plan generator + parent dashboard	This is a core demo moment
38–48	EduScore computation + score band logic + admission-deposit suggestion	Second core demo moment
48–58	FeePassport: signed record export/import between two seeded “schools” in the demo	Even a simulated second school works for the demo
58–66	Cash-flow/runway dashboard + academic performance/achievements tab on parent & admin dashboards	Both are supporting visuals reusing existing dashboard shell, not deep features
66–end	Polish, seed realistic demo data, record a fallback demo video	Always have an offline fallback

Judgment call: don't try to build the risk model, EduScore, FeePassport, payment gateway, and forecasting all to equal depth in the time available. Pick risk-scoring + flexible payment as the one deep, fully working, end-to-end loop, demo EduScore + FeePassport as the second working loop (even between two seeded demo schools), and keep the cash-flow dashboard as a supporting visual rather than a core feature.

13. Differentiation vs PaperBuddy & Typical Competitors

Capability	PaperBuddy / typical ERP	Vidya360
Fee collection & reminders	Yes	Yes (assumed layer, not rebuilt)
Default prediction	No	Yes — explainable risk score
Flexible/EMI payment plans	Rare, manual	Auto-generated from risk score
Cross-school data portability	No	Yes — FeePassport
Portable payment credit score	No	Yes — EduScore
Institution cash-flow forecasting	No	Yes
Unified fee + academic performance view	No (separate systems, if at all)	Yes — results & achievements alongside dues

14. Pitch Narrative / Judging Angle

- Open with the real problem: families don't default because they're unwilling, they default because lump sums and lack of portable history make every school transition a reset to zero trust.
- Position Vidya360 as the fintech layer schools are missing, not a competing ERP — this directly matches the hackathon's FinTech Innovation framing and avoids looking like an attempt to out-build the sponsor's own product.
- Demo order: show a family with a strong EduScore transferring schools — FeePassport carries their history, the new school's dashboard shows the score and immediately offers a reduced deposit instead of a blanket upfront demand. That single flow tells the whole story in under two minutes.
- Close on the safeguard framing: this is a system that removes financial friction for reliable families, never one that blocks anyone — judges will ask about fairness and bias, so having the answer ready (Section 8's safeguards) matters as much as the build.

15. Selected Roadmap Features — Deep Dive

The following five features were selected from the broader roadmap for deeper specification. Each is written to correlate directly with a pillar already built, so it reads as a natural extension rather than scope creep.

14.1 Scholarship / Subsidy Auto-Matching

Correlates with: Defaulter Risk Scoring + Flexible Payment Plans. Where EMI treats a default risk after it's flagged, this pillar intervenes earlier — by checking whether a family is simply unaware of aid they already qualify for.

- Maintains a database of government (state/central scholarship portals, RTE reimbursements) and institution-specific subsidy schemes, each with structured eligibility rules (income bracket, category, gender, class, sibling count, prior academic performance).
- On every risk-score recompute, the same family attributes already used for risk scoring (family fee load, historical payment strain, sibling count) are re-run against the eligibility rule set — no new data collection required.
- High-risk + eligible family → the parent dashboard proactively surfaces “you may qualify for X, apply here” before a due date is missed, rather than only after a lapse.
- School admin view shows aggregate “unclaimed aid” across its at-risk families — a powerful single number for a principal (“₹12L in aid your families haven't claimed”).

Build layer	What it needs	Reuse
Eligibility rule engine	Simple rules table (JSON conditions per scheme)	New, but lightweight — a rules evaluator, not a model
Trigger	Hook into existing risk-score recompute job	Full reuse of Section 4's scoring cycle
Surface	Card on parent dashboard + admin aggregate widget	Reuse existing dashboard components

Safeguard: this feature only ever suggests and links out to the official application — Vidya360 does not submit government applications on a family's behalf in v1, avoiding liability for third-party approval decisions.

14.2 Auto-Clearance / No-Dues Certificate

Correlates with: FeePassport. Once a student's full fee-payment record is already tracked and signed for portability, generating the exit paperwork from that same record is close to free — but it closes the loop on “what happens across a student's entire journey,” which is a strong narrative point in judging.

- Trigger: parent or school marks a student as “exiting” (transfer, graduation, or withdrawal).
- System checks the FeeInvoice/Payment tables for any outstanding dues; if clear, it auto-generates a No-Dues Certificate and (where applicable) a Transfer Certificate pre-filled with enrollment dates, class, and attendance summary already stored in the Student entity.
- If dues are outstanding, the certificate is withheld and the parent dashboard shows exactly what remains — removing the current back-and-forth of “why haven't I received my TC yet” that most Indian schools handle manually and slowly.
- The certificate itself becomes part of the FeePassport payload, so the new school receives verified proof of clean exit alongside the payment history and EduScore.

Build cost: a PDF template + one conditional check against data already modeled in Section 11 — essentially a reporting feature, not a new subsystem, which makes it a good “cheap but impressive” addition to demo.

14.3 Household Aggregation

Correlates with: Risk Scoring's existing family/sibling clustering. Today that clustering only looks at siblings within one school; this extends it across schools for families with children enrolled in more than one institution.

- A Family entity (already implied by familyId in Section 11) is extended to link children across different currentSchoolId values, not just within one school.
- Parent dashboard aggregates all children's dues — across schools — into one combined view: total due this term, combined risk exposure, and a single blended EMI option that spans multiple institutions' invoices.
- Each school still owns and controls its own invoice and payment records; Vidya360 only aggregates the view and coordinates the combined payment schedule — it does not move money between schools' accounts, avoiding settlement complexity.
- Risk scoring benefits too: a family managing three children's fees across two schools is a materially different risk profile than one child at one school, and the score should reflect combined load, not just the single-school view.

Build note: this is a data-modeling extension (cross-school familyId linkage) plus a dashboard aggregation view — no new payment infrastructure required, making it realistic to demo with two seeded schools sharing one family.

14.4 Fraud / Anomaly Detection

Correlates with: Defaulter Risk Scoring. Same underlying signal pipeline (payment timing, amounts, documents, communication), pointed at a different label — not “will they default” but “is this transaction or claim legitimate.”

- Duplicate discount/scholarship claims: flags if the same discount code or scholarship category is claimed by a family whose profile doesn't match prior approved claims, or claimed twice for the same term.

- Payment proof anomalies: for manual payment uploads (bank transfer screenshots, cheque images), a lightweight check flags mismatched amounts, reused images (hash comparison against previously submitted proofs), or edited-looking metadata — a full forensic image-analysis model is out of scope for a hackathon, so keep this heuristic and explainable.
- Chargeback / reversal patterns: flags families with repeated payment-then-reversal cycles, which existing fee tools don't track at all today.
- Every flag routes to a human admin review queue — the system never auto-blocks a payment or auto-penalizes a family; it only surfaces the anomaly for a person to check, which keeps false positives from becoming real harm.

Build note: reuses the same data pipeline as Section 4's risk model (payments, invoices, family clustering); the “new” work is a small set of rule-based checks plus an admin review queue UI, not a new ML model.

14.5 Fee Insurance / Protection Plan

Correlates with: Risk Scoring output as a pricing input. This turns the risk score from a purely defensive signal into something with a monetizable, partner-facing use — relevant if judges ask “what's the business model beyond the school SaaS fee.”

- Structure: an opt-in add-on where a family pays a small per-term premium; if a defined trigger occurs (job loss, hospitalization, death of the fee-paying guardian, documented via a partner insurer's existing claims process), one term's fee is covered directly.
- Vidya360's role is data, not underwriting: the risk score and payment history feed an actual insurance partner's pricing engine as an alternate-data signal, the same way telematics data feeds car insurance pricing. Vidya360 does not itself act as the insurer or hold the risk.
- For the hackathon demo: show the integration point (a premium quote returned from a mocked insurer API call, priced off the family's existing risk score) rather than building real underwriting — the judging value is in showing the model's output has a second real-world consumer, not in building an insurance product from scratch.

Framing note: keep this pitched clearly as future-partnership scope on the roadmap slide rather than a built feature, since actual insurance underwriting requires regulatory partnership that's out of scope for a hackathon build.

16. Next Steps

From here, the two concrete next steps are: (1) scaffold the actual Prisma schema and NestJS module structure for FeePassport and EduScore specifically, since they're the new, unproven parts of the build; or (2) draft the pitch deck narrative and slide structure so the story is locked in before the team splits up to build. Either can be done next.